

Team Biweekly Report Summary

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Period: April 21 - May 11, 2025

Progress Overview

The team has made substantial progress during this reporting period, successfully transitioning from design to development phase and continuing through integration testing. Core backend services and frontend interfaces have been implemented, with the AI model training and validation now complete. Integration between components has proceeded well, with core system integration testing completed according to schedule. The team is on track with the project timeline, with several components progressing ahead of schedule despite the holiday break between April 28 and May 4.

Project Progress Status

According to the project Gantt chart, during this reporting period (April 21 - May 11, 2025), the team has completed or made significant progress on the following WBS tasks:

- Task 4.3.1 (Core Function Development with Unit System): 100% complete (Tong Li)
- Task 4.4.1 (Core Interface Development): 100% complete (Fubin Chen)
- Task 4.5 (AI Model Training): 100% complete (Xuanhe Yang)
- Task 5.3 (AI Model Validation): 100% complete (Xuanhe Yang)
- Task 4.3.2 (Non-Core Function Development with Unit Test): 100% complete (Jingxiao Han)
- Task 4.4.2 (Non-Core Interface Development): 85% complete (Fubin Chen)
- Task 5.1 (Core System Integration Testing): 100% complete (Jingxiao Han)
- Task 5.2 (System Testing): 10% complete (Tong Li)

Current project progress is aligned with the schedule, with several components progressing ahead of the planned timeline. The team has successfully implemented both core and non-core functionalities and completed integration testing, maintaining good momentum across all workstreams despite the holiday break.

Team Contributions in Relation to WBS

- **Xuanhe Yang (32 hours):**
 - WBS 4.5: AI Model Training (6 hours) - 100% Complete
 - WBS 5.3: AI Model Validation (20 hours) - 100% Complete
 - Supporting Core System Integration and Testing (6 hours)
- **Fubin Chen (36 hours):**
 - WBS 4.4.1: Core Interface Development (10 hours) - 100% Complete
 - WBS 4.4.2: Non-Core Interface Development (26 hours) - 85% Complete
- **Tong Li (52 hours):**
 - WBS 4.3.1: Core Function Development with Unit System (40 hours) - 100% Complete
 - WBS 5.2: System Testing (12 hours) - 10% Complete
- **Jingxiao Han (56 hours):**
 - WBS 4.3.2: Non-Core Function Development with Unit Test (28 hours) - 100% Complete
 - WBS 5.1: Core System Integration Testing (28 hours) - 100% Complete

Key Deliverables

1. **Core Backend Services:**
 - Model Management System with import/select/list functionalities
 - User Management Module with authentication and authorization
 - Forecast Analysis System with health status prediction
 - Device Management Module with device selection and listing

2. **Core & Non-Core Frontend Interfaces:**
 - Responsive UI components for all system modules
 - Interactive dashboards with data visualization
 - User management screens with role-based access controls
 - Model management interface with upload/selection capabilities
 - Monitoring Center interface with real-time data visualization
 - Alert System with notification management
 - Report Generation interface with export functionality
3. **AI Model Components:**
 - Fully trained skip-GANomaly model for anomaly detection
 - Completed CNN-LSTM model for remaining useful life prediction
 - Comprehensive validation results with performance metrics
 - Model deployment configuration
4. **System Integration:**
 - API integration between frontend and backend services
 - Data flow pipelines between AI models and web application
 - Authentication service integration
 - Core database integration with application services
 - Integration test suite with automated test cases

Challenges & Solutions

- **Challenge:** Optimizing AI model performance while maintaining reasonable inference times
 - **Solution:** Implemented model quantization and pruning techniques to reduce computational requirements
- **Challenge:** Ensuring consistent data flow between frontend and backend components
 - **Solution:** Established standardized data transfer objects and implemented comprehensive API validation
- **Challenge:** Resolving cross-browser compatibility issues in frontend interfaces
 - **Solution:** Implemented progressive enhancement approach with polyfills for older browsers
- **Challenge:** Managing database performance under high-volume write operations
 - **Solution:** Implemented write buffering and batch processing for sensor data streams
- **Challenge:** Maintaining project momentum during and after holiday break
 - **Solution:** Implemented clear handover documentation and structured task priorities for post-holiday continuation

Next Steps

1. Complete non-core interface development
2. Advance system testing to completion
3. Begin user acceptance testing preparation
4. Prepare deployment environment

Resource Summary

- **Total Working Hours:** 176 hours
- **Comparison to Plan:** The team has utilized approximately 56% of the total project hours (549 hours)
- **Schedule Status:** On track with several tasks ahead of schedule, all other tasks on schedule
- **Project Health:** Overall health is excellent, with progress exceeding expectations in core system components

Biweekly Report

Name: Xuanhe Yang

Period: April 21 - May 11, 2025

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Week 1: AI Model Training Completion (April 21-27, 2025)

During this week, I focused on completing the AI model training and beginning validation procedures. My primary contributions were:

1. **AI Model Training** (April 21-27)
 - Completed training of skip-GANomaly model for anomaly detection
 - Finalized CNN-LSTM model training for RUL prediction
 - Optimized hyperparameters based on validation performance
 - Implemented model serialization for deployment

Time spent: 15 hours

Week 2: Model Validation and Integration Support (May 5-11, 2025)

After the holiday break, this week was dedicated to completing model validation and supporting system testing:

1. **AI Model Validation** (May 5-11)
 - Completed comprehensive validation of anomaly detection model against test dataset
 - Finalized RUL prediction model performance evaluation using RMSE and MAE metrics
 - Created detailed performance visualization dashboards for model evaluation
 - Documented model capabilities and limitations
 - Prepared validation summary report with benchmarking against industry standards
2. **Integration Support**
 - Provided technical support for model deployment in the integrated environment
 - Assisted with troubleshooting inference API issues
 - Created benchmark tests for model performance under production conditions

Time spent: 17 hours

Completed WBS Items

- 4.5 AI Model Training (WBS Item, 12 hours) - 100% Complete
- 5.3 AI Model Validation (WBS Item, 10 hours) - 100% Complete

Challenges & Solutions

The primary challenge during this period was balancing model accuracy with inference performance. Initial validation showed that while our models achieved high accuracy, they required significant computational resources for real-time inference. I addressed this by:

1. Implementing model quantization to reduce memory footprint
2. Applying model pruning techniques to eliminate redundant parameters
3. Designing a batched inference strategy for more efficient processing

Next Steps

1. Support system testing with model-specific test cases
2. Prepare for user acceptance testing
3. Create explanatory materials for model outputs
4. Finalize model deployment documentation
5. Develop maintenance plan for model updates

Total Hours Worked

Total hours for this reporting period: 32 hours

Biweekly Report

Name: Jingxiao Han

Period: April 21 - May 11, 2025

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Week 1: Non-Core Function Development Initiation (April 21-27, 2025)

During this week, I began implementing the non-core functions of our predictive maintenance platform. My specific contributions included:

1. **Monitoring Center Development** (April 21-27)
 - Implemented real-time device health status monitoring functionality
 - Created energy consumption monitoring components
 - Developed parameter visualization mechanisms
 - Established data refresh mechanisms and caching strategy
2. **Alert System Development** (April 25-27)
 - Created alert viewing interface with filtering capabilities
 - Implemented alert confirmation workflow
 - Developed alert categorization and prioritization logic

Time spent: 20 hours

Week 2: Integration Testing and Function Completion (May 5-11, 2025)

After the holiday break, this week was dedicated to completing non-core function development and conducting comprehensive integration testing:

1. **Non-Core Function Completion** (May 5-7)
 - Completed Report Generation Module with template system
 - Finalized alert management functionality with notification services
 - Implemented data export features with multiple format support
 - Added final unit tests and documentation
2. **Core System Integration Testing** (May 7-11)
 - Executed comprehensive integration test suite covering all system interfaces
 - Verified API compatibility and data flow between components
 - Tested authentication and authorization across system boundaries

Time spent: 36 hours

Completed WBS Items

- 4.3.2 Non-Core Function Development with Unit Test (WBS Item, 28 hours) - 100% Complete
- 5.1 Core System Integration Testing (WBS Item, 28 hours) - 100% Complete

Challenges & Solutions

The main challenge I faced was implementing efficient real-time data monitoring while maintaining acceptable system performance. This required balancing update frequency with server load. I addressed this by:

1. Implementing a smart polling mechanism that adjusts frequency based on data volatility
2. Creating a layered caching strategy to reduce unnecessary data transfers

Next Steps

1. Support system testing activities
2. Assist with user acceptance test preparation
3. Develop deployment verification tests

Total Hours Worked

Total hours for this reporting period: 56 hours

Biweekly Report

Name: Fubin Chen

Period: April 21 - May 11, 2025

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Week 1: Core Interface Development (April 21-27, 2025)

During this period, I focused on implementing the core interface components for our ML-based device lifespan prediction web application. I delivered the following components:

- **Model Management System Interface**
- **User Management Interface**
- **Forecast Analysis Interface**
- **Device Management Interface**

Time spent on Core Interface Development: 10 hours

Week 2: Non-Core Interface Development (May 5-11, 2025)

After the holiday break, this week I focused on implementing the non-core interface components:

1. **Monitoring Center Interface**
 - Implemented real-time dashboard with equipment status visualization
 - Created parameter monitoring panels with threshold indicators
 - Developed trend analysis charts with customizable time ranges
 - Implemented equipment health score visualizations
2. **Alert System Interface**
 - Created alert notification center with priority-based sorting
 - Implemented alert detail view with action tracking
 - Developed alert history and resolution workflow
 - Created alert settings configuration interface
3. **Report Generation Interface**
 - Implemented report template selection and customization
 - Created data visualization components for reports
 - Developed export functionality with multiple format support
 - Implemented scheduled report configuration

Time spent: 26 hours

Completed WBS Items

- 4.4.1 Core Interface Development (WBS Item, 28 hours) - 100% Complete
- 4.4.2 Non-Core Interface Development (WBS Item, 26 hours) - 85% Complete

Challenges & Solutions

The main challenge I faced was ensuring consistent user experience across different browsers and device types. I addressed this by:

1. Implementing a responsive design system based on fluid grid layouts
2. Creating device-specific optimizations for critical interfaces

Next Steps

1. Complete remaining non-core interface components
2. Finalize styling and visual consistency across all interfaces

Total Hours Worked

Total hours for this reporting period: 36 hours

Biweekly Report

Name: Tong Li

Period: April 21 - May 11, 2025

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Week 1: Core Function Development (April 21-27, 2025)

During this week, I focused on implementing the core backend services for our predictive maintenance platform.

My specific contributions included:

1. **Model Management System Development** (April 21-24)
2. **User Management Module** (April 24-27)
3. **Forecast Analysis System Development** (April 26-27) Time spent: 40 hours

Week 2: System Testing Preparation (May 5-11, 2025)

After the holiday break, this week was dedicated to preparing and beginning system testing:

1. **System Testing Preparation and Execution** (May 5-11)
 - Developed comprehensive system test plan with test scenarios
 - Created automated test scripts for core functionality validation
 - Implemented performance testing framework for load testing
 - Conducted initial system tests focusing on core functionalities
 - Documented test results and identified optimization opportunities
2. **Database Performance Optimization**
 - Implemented database query optimization for high-volume operations
 - Created additional indexes for frequently accessed data patterns
 - Developed database maintenance procedures for production environment

Time spent: 12 hours

Completed WBS Items

- 4.3.1 Core Function Development with Unit System (WBS Item, 40 hours) - 100% Complete
- 5.2 System Testing (WBS Item, 12 hours) - 10% Complete

Challenges & Solutions

The primary challenge I faced was designing an efficient system for handling high-volume sensor data while maintaining low-latency access for analytical queries. I addressed this by:

1. Implementing a multi-tier data storage architecture with hot/warm/cold data zones
2. Creating specialized indexes for time-series data with optimized compression
3. Developing a smart data aggregation system that pre-calculates common metrics

Next Steps

1. Continue system testing with expanded test scenarios
2. Address any issues identified during testing
3. Prepare for user acceptance testing
4. Finalize database optimization for production deployment
5. Create operational documentation for database management

Total Hours Worked

Total hours for this reporting period: 52 hours